

On roots near the unit circle

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Abstract

We define n -exteriority of an algebraic integer—the number of its Galois conjugates outside the unit circle. Although this is ubiquitous in the literature, it has not been treated as a named invariant. We isolate it and prove bounds on crowding near roots of unity that force the exterior gap ϵ away from zero.

1 n -Exteriority

Definition 1.1. Let α be an algebraic integer with Galois conjugates $\{\sigma(\alpha)\}$ over $\mathbb{Q}$. The *exteriority* of α is

$$\#(\alpha) := |\{\sigma : |\sigma(\alpha)| > 1\}|$$

i.e. the number of conjugates of α lying strictly outside the unit circle (counted with multiplicity).

If $P \in \mathbb{Z}[x]$ is a monic irreducible polynomial and α is a root of P , we set

$$\#(P) := \#(\alpha)$$

which is well-defined.

Definition 1.2. We are interested in the set of polynomials,

$$S(n, \epsilon) = \{P : \#(P) = n, \ 0 < \rho_1 - 1 \leq \epsilon\} \quad (1)$$

where $P \in \mathbb{Z}[x]$ are monic, irreducible, non-cyclotomic, reciprocal polynomial. We call ϵ the *exterior gap*, and $1 < \rho_1 \leq \rho_2 \leq \dots \leq \rho_{\#(P)}$ are the moduli outside the unit circle (with $\rho_1 = \min_{|\alpha_j| > 1} |\alpha_j|$, and therefore $\rho_{\#(P)}$ is the house of P). Throughout this paper we will loosen notation and write $\rho := \rho_{\#(P)}$ for the house, unless an index is specified.

Remark 1.3. With this we have a simple definition for the set of Salem polynomials when the degree $d = \deg P$ is even and $d \geq 4$

$$\bigcup_{\epsilon > 0} S(1, \epsilon) = S(1, \infty) = \{\text{Salem polynomials}\}$$

Lemma 1.4. Since $M(P) = \prod_{i=1}^n \rho_i$ and each $\rho_i \geq \rho_1 = 1 + \epsilon$, we get

$$M(P) \geq (1 + \epsilon)^n$$

Remark 1.5. Consider 2-exterior monic, irreducible, non-cyclotomic, reciprocal polynomials. These are the elements of $P \in S(2, \epsilon)$. These behave very similarly to Salem polynomials, only differing by an extra root outside of the unit circle. For this set we have

$$m(P) = \ln(\rho_{\#(P)}) + O(\epsilon)$$

Definition 1.6. Define the symmetric wedges $W(\zeta; \theta_w, \delta_r) := \{z : ||z| - 1| \leq \delta_r, \quad |\text{Arg } z - \text{Arg } \zeta| \leq \theta_w\}$. We call δ_r and θ_w , ‘the radial half-thickness’ and ‘half-angle’ respectively. Furthermore, let $U_q(\theta_w, \delta_r)$ be the union of symmetric wedges centered at the primitive q -th roots of unity, defined as

$$U_q(\theta_w, \delta_r) := \bigcup_{\substack{\zeta \text{ primitive} \\ \zeta^q = 1}} W(\zeta; \theta_w, \delta_r)$$

Let

$$N_q = N_q(P; \theta_w, \delta_r) := |\{\text{conjugates of } P \text{ in } U_q(\theta_w, \delta_r)\}|$$

$$\eta_q := \frac{N_q}{d}$$

Definition 1.7. One may also define the wedges as $W^{\leq 1}(\zeta; \theta_w, \delta_r) := \{z : 1 - \delta_r \leq |z| \leq 1, \quad |\text{Arg } z - \text{Arg } \zeta| \leq \theta_w\}$

$$U_q^{\leq 1}(\theta_w, \delta_r) := \bigcup_{\substack{\zeta \text{ primitive} \\ \zeta^q = 1}} W^{\leq 1}(\zeta; \theta_w, \delta_r)$$

$$N_q^{\leq 1} := N_q^{\leq 1}(P; \theta_w, \delta_r) := |\{\text{conjugates of } P \text{ in } U_q^{\leq 1}(\theta_w, \delta_r)\}|$$

$$\eta_q^{\leq 1} := \frac{N_q^{\leq 1}}{d}.$$

This alternative definition allows one to construct bounds independent of the exterior gap, however, these are slightly weaker than the bounds using Definition 1.6. so we will stick to Definition 1.6.

Theorem 1.8. Let $P \in \mathbb{Z}[x] \in S(n, \epsilon)$ of degree d , and write $\rho = \rho_{\#(P)}$ for its house. Fix $q \geq 1$, a half-angle $\theta_w \in (0, \pi]$, and a radial half-thickness $\delta_r \in (0, 1]$. If $N_q \geq 1$, then

$$\theta_w + \delta_r \geq (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q}} \quad (2)$$

In particular, if $\delta_r \geq \epsilon$, then the exterior gap $\epsilon = \rho_1 - 1$ satisfies

$$\epsilon \geq (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q}} - \theta_w \quad (3)$$

where $\varphi(q)$ is Euler’s totient function.

Proof. For a primitive q -th root ζ and a conjugate α of P , by the triangle inequality and $|1 - e^{ix}| = 2|\sin(x/2)| \leq |x|$,

$$|\zeta - \alpha| \leq |e^{i \text{Arg } \zeta} - e^{i \text{Arg } \alpha}| + ||\alpha| - 1| \leq |\text{Arg } \alpha - \text{Arg } \zeta| + ||\alpha| - 1|$$

Hence, if $\alpha \in W(\zeta; \theta_w, \delta_r)$ then $|\zeta - \alpha| \leq \theta_w + \delta_r$, while in general $|\zeta - \alpha| \leq 1 + |\alpha| \leq 1 + \rho$. Writing $P(z) = \prod_{j=1}^d (z - \alpha_j)$ and letting N_ζ be the number of α_j in $W(\zeta; \theta_w, \delta_r)$, we get

$$|P(\zeta)| = \prod_{j=1}^d |\zeta - \alpha_j| \leq (\theta_w + \delta_r)^{N_\zeta} (1 + \rho)^{d - N_\zeta}$$

Multiply over all primitive ζ with $\zeta^q = 1$. Since P is monic and coprime to Φ_q , the product equals the nonzero integer $\prod_\zeta P(\zeta) = \text{Res}(P, \Phi_q)$, hence its absolute value is ≥ 1 . With $S := \sum_\zeta N_\zeta$,

$$1 \leq \prod_\zeta |P(\zeta)| \leq (\theta_w + \delta_r)^S (1 + \rho)^{\varphi(q)d - S}$$

Rearranging yields

$$\theta_w + \delta_r \geq (1 + \rho)^{1 - \frac{\varphi(q)d}{S}}$$

Since each counted conjugate lies in at least one wedge, $S \geq N_q$, and $x \mapsto (1 + \rho)^{-(\frac{\varphi(q)d}{x} - 1)}$ is increasing for $x > 0$, hence

$$\theta_w + \delta_r \geq (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q}} = (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q}}$$

which is (2). If moreover $\delta_r \geq \rho_1 - 1$, then $\theta_w + \varepsilon \leq \theta_w + \delta_r$, so (2) implies

$$\varepsilon \geq (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q(\delta_r)}} - \theta_w$$

which is (3). □

Remark 1.9. The bound (3) is non-trivial precisely when

$$\eta_q > \frac{\varphi(q)}{1 - \log_{1+\rho}(\theta_w)}$$

In practice one maximizes the RHS over a small set of orders q (e.g. $q = 1, 2, 3, 4, 5$) to obtain the best lower bound for ε .

Corollary 1.10. *Using Definition 1.7. one has*

$$\theta_w + \delta_r \geq (1 + \rho)^{1 - \frac{\varphi(q)}{\eta_q^{\leq 1}}}.$$